

Omar ElMasry

AI Engineer

Contact

Location: Cairo, Egypt

LinkedIn: [linkedin.com/in/omar-ai-demo](https://www.linkedin.com/in/omar-ai-demo)

GitHub: github.com/omasry-ai

Professional Summary

Data Scientist and Machine Learning Engineer with 3+ years of hands-on experience in designing, developing, and deploying advanced machine learning models in production environments. Skilled in extracting business insights from large, complex datasets and collaborating with cross-functional teams to deliver impactful AI solutions. Expert in Python, data analysis, and state-of-the-art model development, with a strong background in NLP, information extraction, model optimization, and AI agent design.

Recognized for strong communication abilities and a passion for driving innovation through data science.

Education

MSc in Data Science

Cairo University, Faculty of Computers & Artificial Intelligence

2018 – 2020

BSc in Computer Science

Helwan University, Egypt

2014 – 2018

Professional Experience

Senior Machine Learning Engineer

EgyData Solutions, Cairo, Egypt

March 2022 – Present

- Lead the end-to-end lifecycle of machine learning projects for multinational clients, focusing on real-time anomaly detection and NLP-based information extraction.
- Designed, built, and deployed scalable classification and recommendation models using Python, scikit-learn, and TensorFlow.
- Collaborated with data engineers, DevOps, and business stakeholders to integrate model outputs into customer-facing web applications, delivering actionable insights and measurable value.
- Mentored junior data scientists, establishing coding and experimentation best practices.

Data Scientist

MinaTech Analytics, Cairo, Egypt

July 2020 – February 2022

- Developed entity recognition pipelines and vector search engines for unstructured document repositories, improving information retrieval efficiency by 35%.
- Conducted systematic data exploration and wrangling of multi-terabyte datasets using pandas and SQL.
- Collaborated with software engineers on productionizing machine learning systems, incorporating MLOps principles for automation and monitoring.
- Communicated modeling results and business impact clearly to non-technical stakeholders through reports and interactive visualizations.

Projects

Automated Legal Document Classifier

- Built an NLP-driven classification system that automatically tags legal contracts based on content and intent using transformers (HuggingFace BERT).
- Integrated entity extraction to identify parties, dates, and key obligations.
- Deployed the solution as a scalable RESTful API, enabling downstream applications to leverage classification and extraction in real time.

Fraud Detection AI Agent

- Designed a machine learning pipeline to detect transaction anomalies and potential fraud in financial data.
- Employed advanced feature engineering, dimensionality reduction, and ensemble learning for model robustness.
- Implemented model monitoring to flag concept drift and retrain models with fresh data.

Semantic Search with Vector Databases

- Developed an unstructured document search solution based on vectorized embeddings (FAISS and spaCy).
- Enabled contextual and semantic querying over millions of text records, boosting retrieval accuracy over traditional keyword search by 42%.
- Deployed the engine within a document management system for a major logistics client.

Technical Skills

Programming: Python, R, Bash

Data Science Tools: pandas, NumPy, scikit-learn, matplotlib, seaborn

Machine Learning: Regression, Classification, Clustering, NLP, Entity Recognition, Ensemble Methods, Deep Learning (TensorFlow, Keras)

NLP: spaCy, NLTK, HuggingFace Transformers

Data: SQL, NoSQL, Spark, Airflow

Deployment: Docker, Flask/FastAPI, AWS EC2/Sagemaker

Version Control: Git, GitHub

Data Visualization: matplotlib, seaborn, Tableau

Search & Databases: FAISS, Elasticsearch

Certifications

- TensorFlow Developer Certificate – TensorFlow (2021)
- IBM Applied Data Science Professional Certificate (Coursera, 2020)

Soft Skills

- Strong written & verbal communication
- Cross-functional team collaboration
- Stakeholder management
- Problem-solving and analytical thinking
- Mentorship and team facilitation
- Agile and adaptive to new technologies